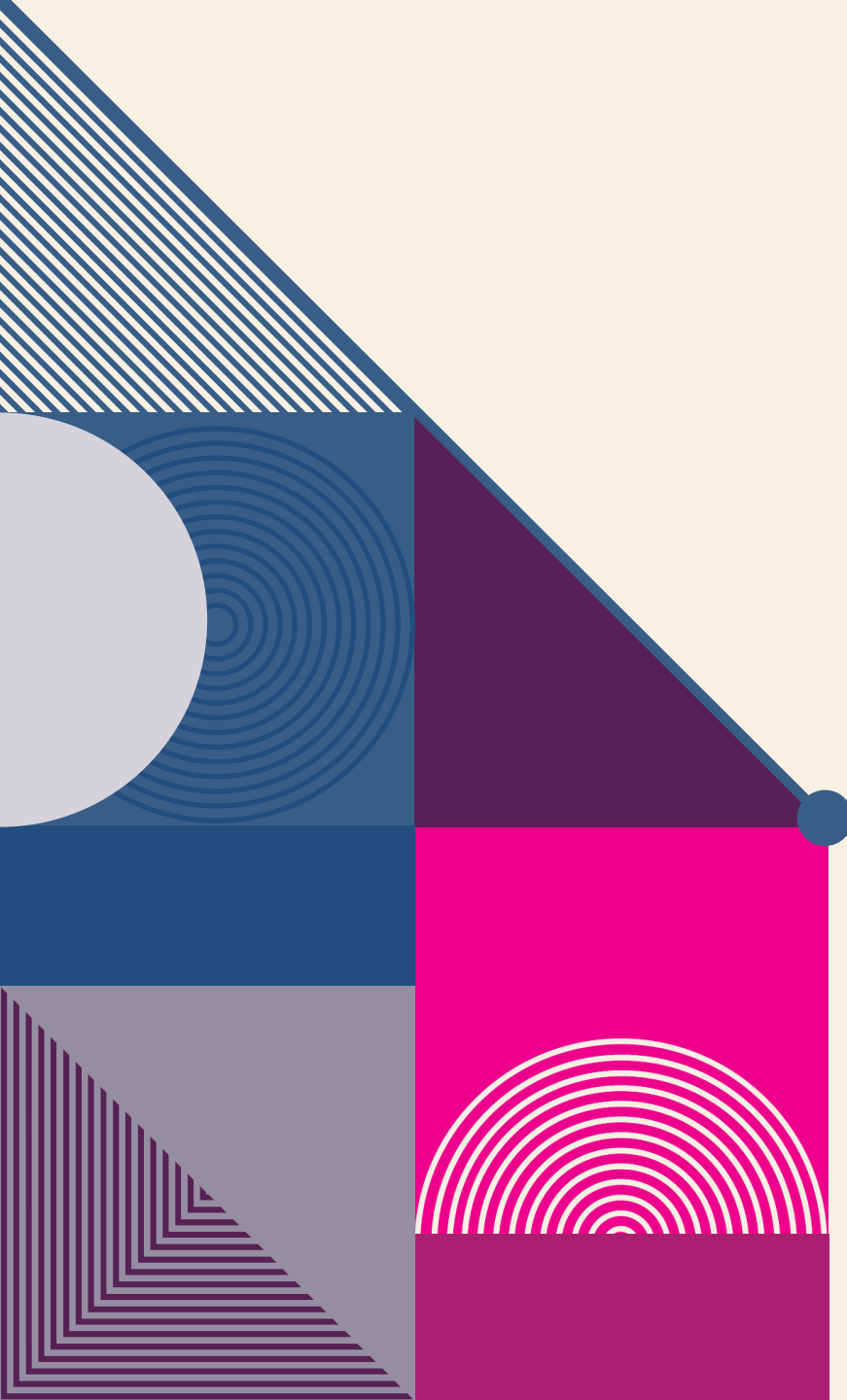


An abstract geometric design on the left side of the slide. It features a dark blue background with various geometric shapes and patterns. A white circle is positioned near the top left. Below it, there are concentric circles in a lighter blue shade. A large, light gray semi-circle is also present. The design is composed of several rectangular and triangular blocks in shades of blue, purple, and pink. Some blocks have diagonal line patterns, while others are solid colors. A white diagonal line runs from the top left towards the bottom right, separating the abstract design from the text area.

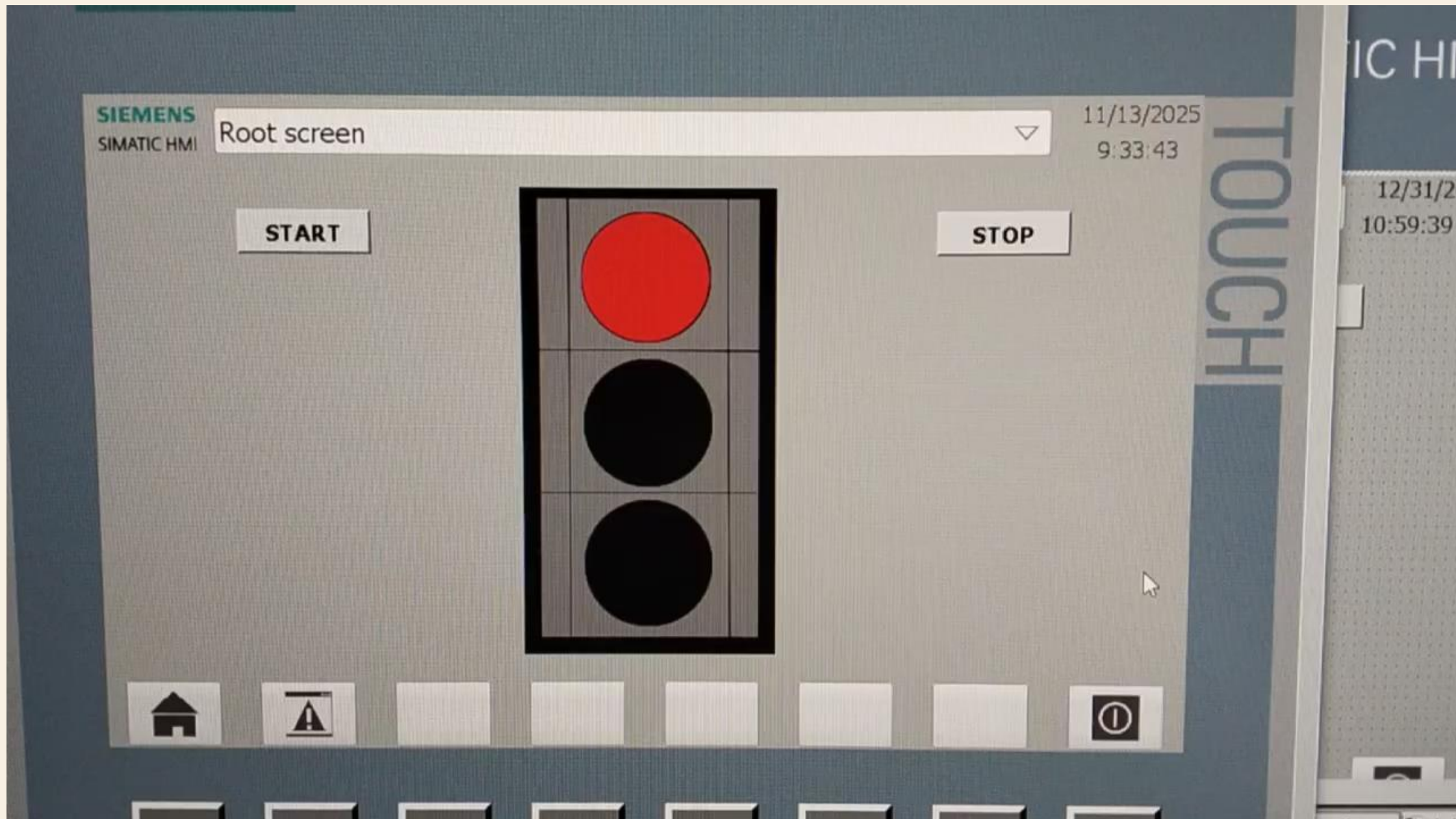
PRACTICE WORK 8

VIDEO



TASK 1

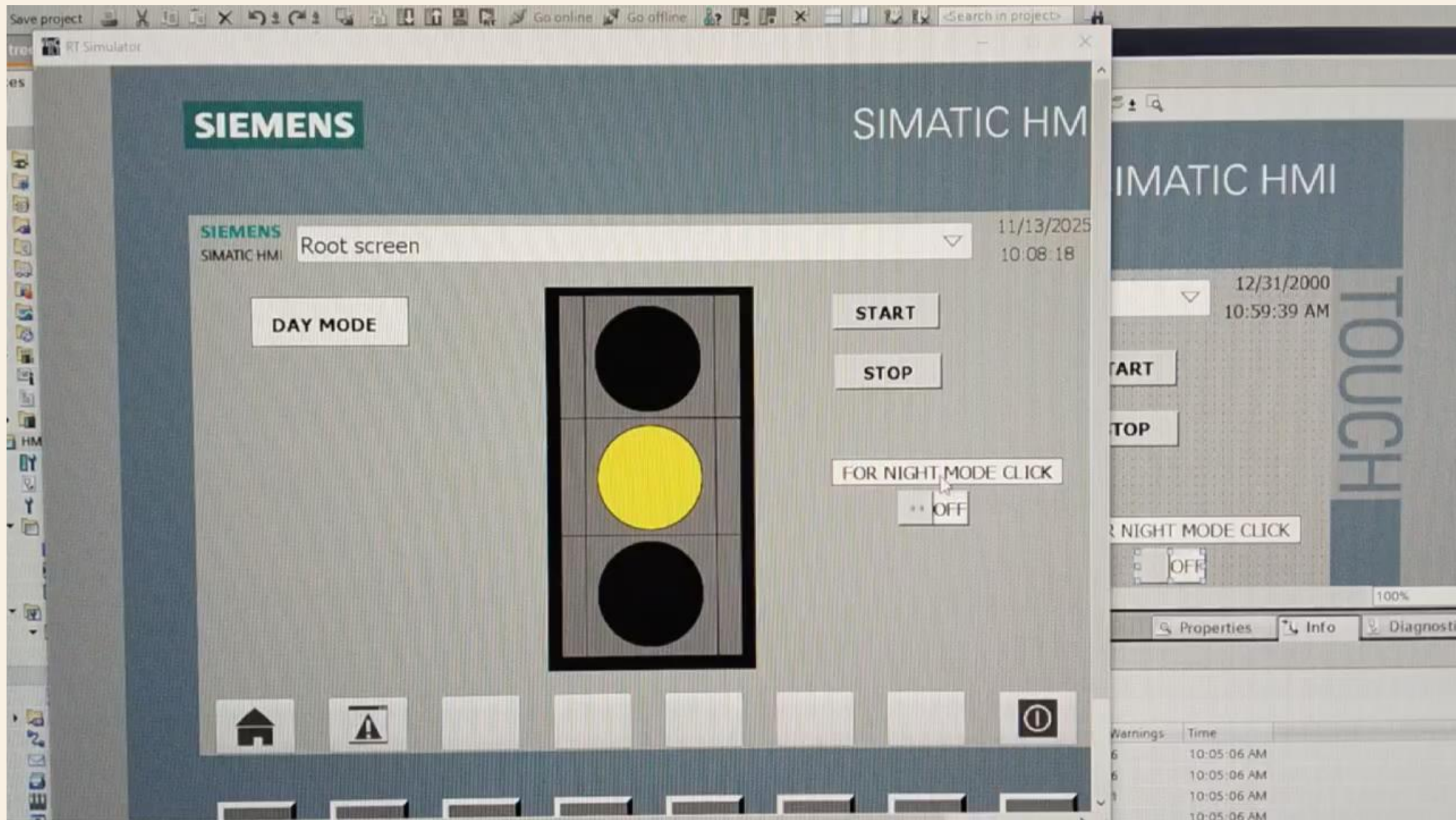
In this task you need to create a program to control traffic lights with three light signals: red, yellow and green. The traffic light operation is implemented using three timers that sequentially activate the corresponding signals. It is also necessary to visualize the traffic light operation on the HMI panel.





TASK 2

Develop improved program traffic light control program, adding the "flashing yellow signal" mode for night time operation. Realize automatic switching between day and night modes by a set time or via a control signal. Update the visualization on the HMI to show both modes of operation.

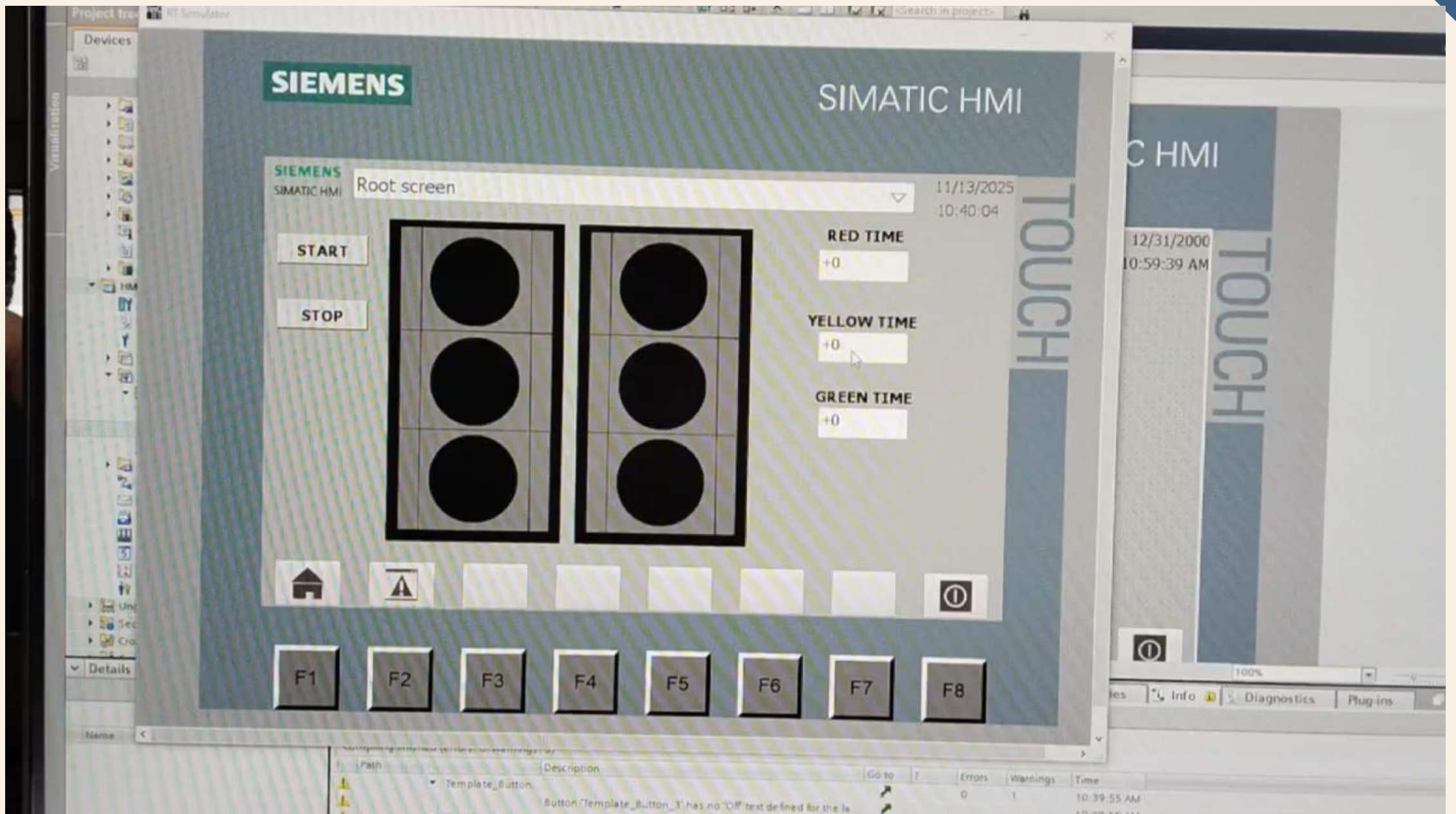


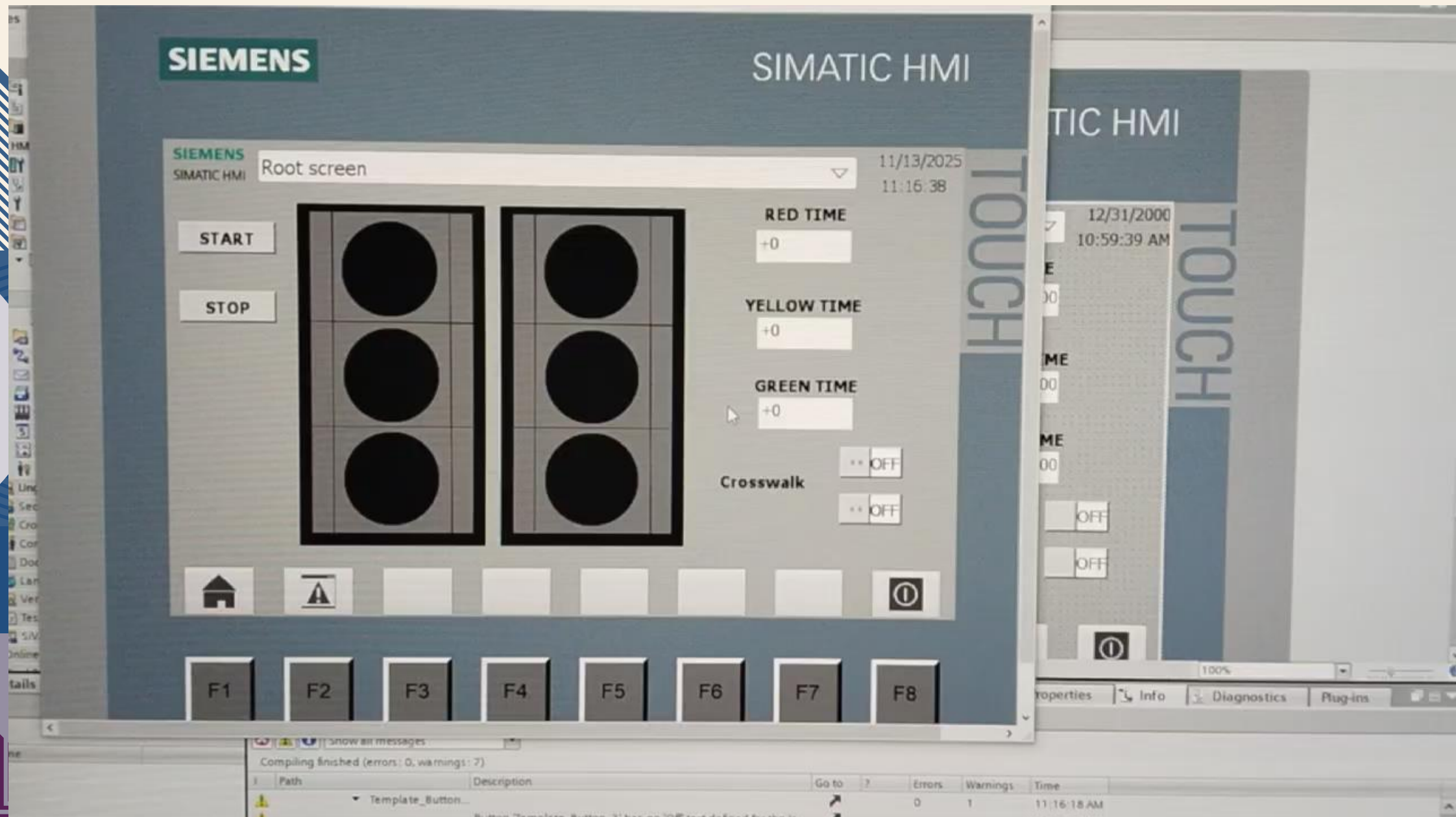


TASK 3

Develop a traffic light control program with three signals (red, yellow, green), where the duration of each signal is set and can be changed from the HMI panel in real time. The switching of states should be done using timers.

Additionally implement the pedestrian enquiry function: to activate it, two pedestrian buttons should be pressed simultaneously, after which the duration of the current signal is reduced to a minimum value.





SIMULATION

